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Palaeomagnetism of the Torridonian of Rhum, Scotland: evidence for limited uplift of the Central Intrusive Complex

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The Tertiary Central Intrusive of Rhum was emplaced into the Torridonian Sandstone, causing a heat influx into the sediment. Detailed progressive demagnetisation of sandstones from the thermal aureole of the intrusion show that they reached maximum temperatures which decrease systematically with distance in a manner indicative of conduction-dominated cooling. Temperatures reached near to the intrusion are higher than would be predicted theoretically, probably as a result of the presence of smaller plugs of magma at depth which are only rarely exposed at the surface. The shape and magnitude of the thermal aureole suggest that upward movement of the Central Intrusive along the bounding fault was limited.

Primary remanence in Torridonian sandstones unaffected by the intrusion is carried by both magnetite and haematite, often with both occurring in the same specimen. It is suspected that only magnetite carries a truly primary (detrital?) remanence; the haematite probably formed diagenetically (largely by oxidation of detrital magnetite) at some time shortly after deposition.

1. Introduction

The island of Rhum in the Inner Hebrides, Scotland, consists of a Tertiary Layered ultrabasic intrusion emplaced along steep ring faults into the Torridonian Sandstone. The chief aim of this study is to investigate the cooling history of the ultrabasic intrusion as recorded by changes in the magnetic characteristics of its aureole. In particular, the width, peak temperature profile and magnetic mineralogy of the aureole may suggest whether cooling was predominantly conductive, convective, or if it had features of both mechanisms. In addition, samples have been analysed from several sites beyond the detectable limit of the aureole, in unaltered Torridonian sandstones.

A total of 97 samples from 19 sites in the aureole have been analysed, sites being collected in a linear traverse approximately perpendicular to the ring fault bounding the central intrusive. Seven samples were also collected in the contact zone of a small Tertiary dyke intruded into the Torridonian.

Progressive thermal demagnetisation has demonstrated the presence of four distinct zones concentrically surrounding the intrusion. The inner zone shows a single component of remanence directed upwards and to the south, presumably of Tertiary age; this was acquired after deformation of the sediments by the intrusion and is carried by both magnetite and haematite.

Between 2 and 2.5 km from the ring fault is a zone where magnetite has been completely remagnetised by the intrusion. This Tertiary remanence has blocking temperatures (T_b) up to 580 °C, while the hematitic high-T_b component retains a Torridonian remanence.

Between 2.8 and 3.5 km, occasional Tertiary components held in magnetite may be observed but these are sparse and have a maximum T_b of less than 580 °C. Some Torridonian components with a T_b of less than 580 °C are observed in this zone but high-T_b Torridonian components are predominant.

Further than 4.5 km from the intrusion, the sediments carry only a Torridonian remanence. This is sometimes multicomponent, with a low-T_b

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component predominantly of SE + polarity and a more widespread high-T_b component usually of NW − polarity. Both NW − and SE + components may sometimes occur in the same specimen. Either blocking temperature range may occur to the exclusion of the other, but both are usually present. These are interpreted as representing a primary syn-depositional (?detrital) remanence and a later diagenetic haematite remanence, acquired over more than one polarity interval. The latter probably results from partial oxidation of primary magnetite and/or the breakdown of detrital ferromagnesian minerals, producing haematite.

It is not the aim of this study to attempt to refine previous palaeomagnetic studies of the Tertiary or of the Torridonian, both of which have been well investigated, but rather to use this earlier work to help to identify the age of magnetic components revealed here.

2. Geology

The sampled sediments come entirely from the Loch nan Eala Arkose member of the Applecross Group [1,2]. A recalculated Rb-Sr age of 788 ± 17 m.y. has been determined for the base of this group [3]. The sediments are quartzo-feldspathic sandstones with accessory detrital biotite, muscovite, epidote, zircon, sphene and iron oxides together with occasional exotic rock fragments and reworked sedimentary clasts. Hematitic staining is ubiquitous apart from in the baked contacts of igneous rocks; cementation is predominantly calcareous. Occasional heavy mineral bands, composed chiefly of specularite/magnetite and zircon, give well-developed load-casts in places. The depositional environment has been interpreted as that of a large braided river system with palaeocurrents from the north or northwest [4]; large-scale grain size variations presumably represent the fluctuating energy of this system.

Soft-sediment deformation within the area is widespread, with occasional well-developed slump structures. There is very little tectonic deformation, however, apart from a regional tilt of approximately 15° to the west-northwest. Emplacement of the central intrusive along the steep bounding ring fault has caused deformation so that the strike of the Torridonian swings round and becomes tangential to the outcrop of the fault.

The main feature of the Tertiary geology of Rhum is a layered ultrabasic intrusion, approximately 7 km in diameter (Fig. 1); this is thought to have been emplaced by upward movement of up to 2000 m along the encircling ring fault [5]. A number of small plugs and dykes are associated with the main intrusion; they extend as far as the northern coastline of the island, where better exposure causes them to appear to be more numerous than closer to the intrusion. The narrowness of most dykes suggests that they were unlikely to have caused a significant regional heat influx.

Oxygen isotope and mineralogical studies have shown igneous intrusions of the British Tertiary Igneous Province (BTIP) to have been the sites of major hydrothermal circulation systems [6–8]. In most of the centres of plutonic activity, the central intrusive rocks were emplaced into a fractured and highly permeable pile of Tertiary lavas. Oxygen isotope values are substantially depleted, both in the intrusions and in the large aureoles surrounding the central complexes, presumably due to interaction with circulating meteoric fluids. However, Rhum is unique among the intrusive centres of the BTIP in that the central complex was intruded into Precambrian Lewisian Gneiss and Torridonian Sandstone, both of which are of very low permeability. $\delta^{18}O$ studies [9] show that significant oxygen isotope depletion on Rhum occurs only in a narrow band (< 0.5 km wide) coincident with the main Ring Fault. This indicates that hydrothermal or meteoric fluids rose along or near to the Ring Fault and that there was no significant hydrothermal circulation within the Torridonian Sandstone.

3. Previous palaeomagnetic studies

3.1. Torridonian

The Torridonian Sandstone has been the subject of extensive palaeomagnetic research; the two early studies [10,11] were among the first palaeomagnetic investigations of ancient sediments. They identified a dominant NW − /SE + magnetisation axis in the Aultbea and Applecross Groups and suggested that the remanence was carried in detrital magnetite/specularite grains, although progressive demagnetisation was not used. Later

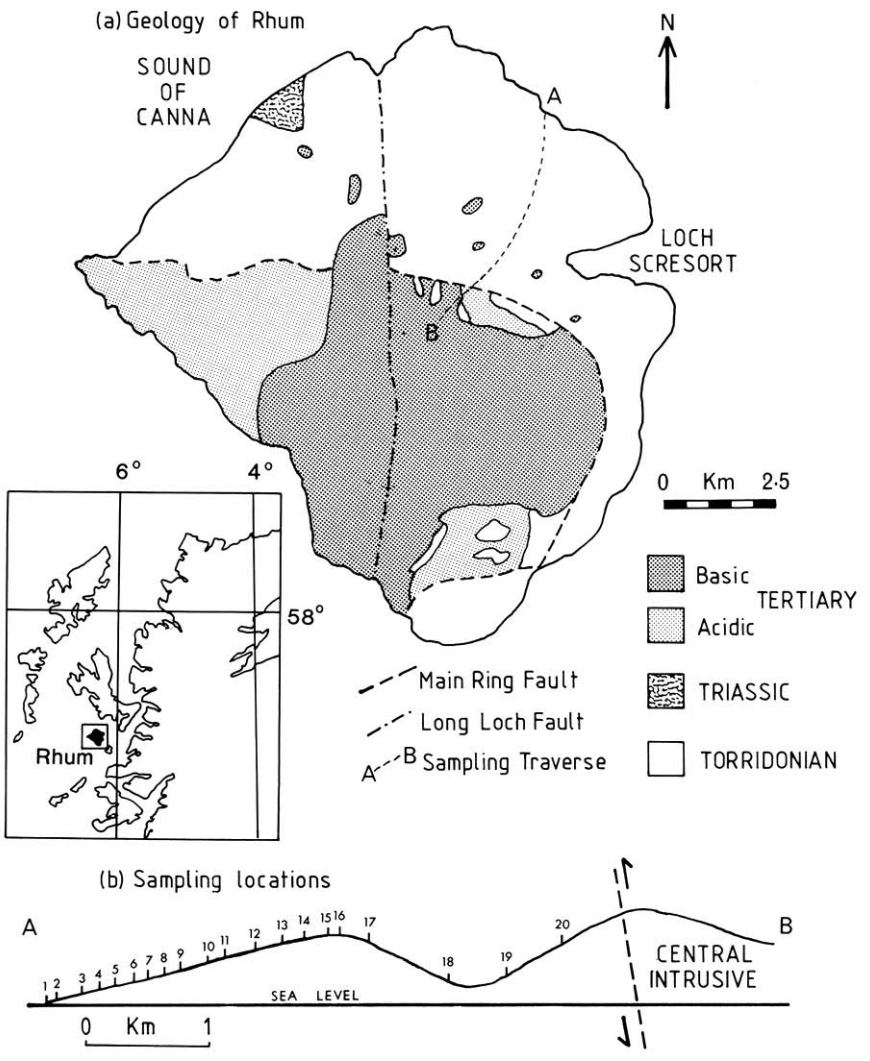


Fig. 1. Geology and sampling locations. (a) Geology of Rhum (simplified). Torridonian sandstones occur both within and outside the ring fault (not differentiated). Sites located along line *A-B*. (b) Section along *A-B* (×2 vertical exaggeration) showing positions of sampling locations. Strike of the Torridonian is perpendicular to the section.

work [12] using thermal cleaning found directions in the Stoer Group which were broadly similar to those found in the earlier studies. It was concluded that the remanence was acquired at the time that the sediments were deposited. The most recent published study [13] confirmed that the magnetic remanence was acquired during and/or shortly after deposition and that the fine-grained red pigment contributed only a small fraction of the total remanence. Specularite grains carry much of the remanence, although magnetite was detected thermomagnetically.

A summary of the magnetic stratigraphy at various locations [13, fig. 4, incorporating earlier results] showed the presence of a number of magnetic reversals in the southern area of the Applecross Group which are not detected further north. This apparent contradiction was interpreted as reflecting imprecision in stratigraphic correlation between one area and another. Published palaeomagnetic results for the Applecross Group are given in Table 1.

3.2. Tertiary

Palaeomagnetism of the BTIP has also been extensively investigated; full references are given

TABLE 1

Previous work: lower Torridon Group

Group/Formation	D (°)	I (°)	N	k	α_{95} (°)	Reference
Torridon Group (SE+)	129	51	53	14.7	5	[10]
(NW−)	294	−28	28	9.4	9	[10]
Basal Torridon Group	106	61	11	10.0	16	[12]
Applecross Formation	092.7	68.2	17	12.1	10.7	[13]
	124.2	67.0	18	13.7	9.7	[13]
	106.5	56.4	10	13	13.7	[13]

All unit weighting to sites.

in [14]. The main finding of these studies is that emplacement of the central intrusive complexes occurred over more than one polarity interval, both normal and reversed polarities often being found in the same body. The central complex of Rhum has, however, been shown to be exclusively of reversed polarity, as have the extrusive lavas associated with it. Most Tertiary dykes (typically 73–92% in any one area) are also of reversed polarity.

The Central Intrusive of Rhum has a mean direction of approximately $D = 177°$, $I = -58°$ [14,15] although the former study suggests somewhat steeper inclinations of 67–72° in nearby lavas and dykes. A normal polarity remanence was found only in three of the four dykes cutting lavas; all of the 57 coastal dykes examined were found to be of reversed polarity.

4. Sampling and treatment

In this study, samples were collected from a total of 20 sites in the Torridonian Sandstone, in an approximately linear traverse perpendicular to the ring fault bounding the central intrusive (Fig. 1). With the exception of site 16, collected in the contact zone of a Tertiary dyke, sampling locations were chosen to be as far as possible from any known minor Tertiary igneous bodies. Sites are serialised with higher site numbers (up to 20) being successively nearer to the Ring Fault. Site 1 is 5.2 km distant, the remainder having an average separation of 280–300 m. There is little lithological variation between sites, although some samples from heavy mineral accumulations have a high incidence of opaque grains. Small-scale tectonic disturbance is minimal. Between 5 and 8 cores were collected from each site, orientated by a combination of magnetic and solar compass wherever possible, giving an orientation accuracy of approximately ±1°. Greater errors may occur in highly baked, brittle rocks when cores may have a tendency to break off prior to orientation; difficulties in realignment may explain the few totally inconsistent component directions revealed during treatment.

All of the samples discussed in this paper were treated by progressive thermal demagnetisation, typically comprising a total of between 6 and 15 steps; demagnetisation was continued until magnetic behaviour became erratic and uninterpretable. Measurements were made on a Digico spinner magnetometer or a CCL cryogenic magnetometer. Susceptibility, monitored on selected samples throughout treatment, showed little or no change. A small number of samples were demagnetised chemically, using HCl, and by alternating field. Both methods resulted in highly erratic, geologically uninterpretable behaviour and will not be considered further.

Directional data from each specimen were analysed using a line-fitting program (LINEFIND) which uses the algorithm given by [16].

5. Palaeomagnetic results

5.1. Torridonian

A suspected Torridonian remanence (see discussion above) has been found in one or more specimens at every site between 1 and 17, with the exceptions of three (8, 9 and 14) which are dominated by an isothermal remanence (IRM). These sites are characterised by high NRM intensity, single-component behaviour and random in-

tra-site directional distribution: such components will not be considered further.

Two discrete Curie Points (T_c) may be seen, one at 560–585 °C, the other at 670–700 °C. It is suspected that these correspond to magnetite and haematite respectively. Either or both may be found in the same specimen.

Some specimens show a very clear multi-component remanence, with the two T_b ranges being, fortuitously, of opposing polarity. One of the clearest examples of this is shown in Fig. 2a, where the sharp change in polarity at around the T_c of magnetite is apparent.

The lower-T_b component is, however, generally less well-defined than the higher; a more typical example is shown in Fig. 2b. A definable low-T_b Torridonian component is found in 8 sites only, and in every case it is of SE + polarity. Elsewhere, a discrete low-T_b component can often be observed but it is of random orientation even though the high-T_b component may show good directional grouping. Such sites tend also to have at least one specimen showing a complete IRM, suggesting that elsewhere within the site partial isothermal remagnetisation of low-viscosity magnetite may have occurred, presumably at the outer limit of the effects of a lightning strike [17].

Higher-T_b (hematitic) components are generally better defined and more widespread than low-T_b components, and may sometimes be the only stable ancient component present (e.g. Fig. 2c). They may be found as near to the intrusion as site 17; they are also observed at each site further from the intrusion than this (with the exceptions of sites showing a total IRM and also site 16 which is in close proximity to a dyke). Occasional

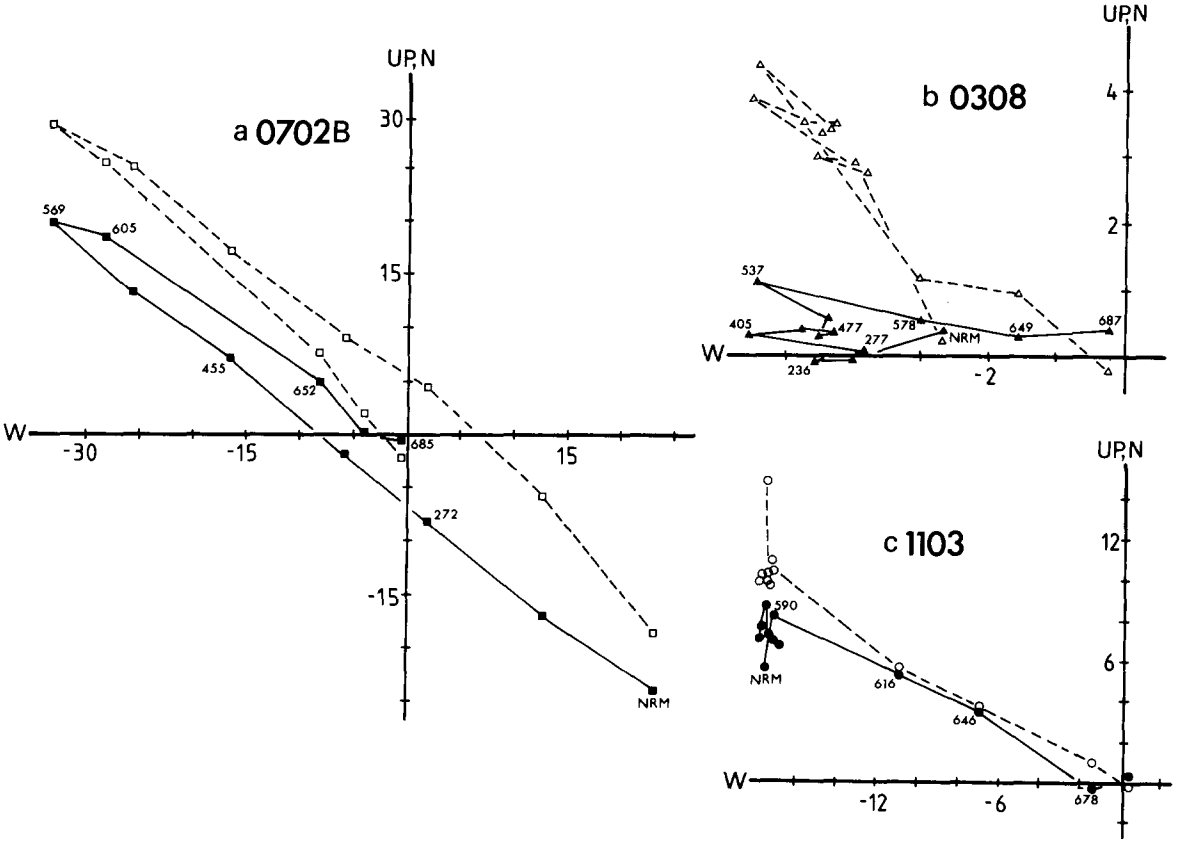


Fig. 2. Torridonian Sandstone: Progressive thermal demagnetisation (orthogonal projections). Demagnetisation temperatures in °C; axes in mA/m. Closed/open symbols and solid/dashed lines represent the horizontal/vertical projections respectively. Note that discussion of low-T_b (recent) components is not given in figure captions. (a) 0702B. Two-component Torridonian remanence; both components well-defined and of opposite polarity. Note high NRM intensity. (b) 0308. As before but low-T_b component is less well-defined. (c) Single-component Torridonian remanence; insignificant intensity decrease below 590 °C.

specimens may show a multi-component high-T_b remanence (e.g. Figs. 3b, 6b).

Site and formation mean directions for both low- and high-T_b components are given in Table 2. Mean directions can be seen to be in good agreement with previously-published studies (Table 1).

Site numbers 1 to 15 correlate directly with relative stratigraphic position (since sites were collected at steadily increasing height above sea level in a direction parallel to the bed strike direction of the sandstones). From this (Table 2), it may be observed that there is no clear magnetic stratigraphy; the two polarities occur in close association, often within the same site.

Origin of the Torridonian NRM. It would appear from the polarity contrast between the low- and high-T_b components in the same specimen (see above) that there was a significant period, spanning at least one field reversal, between their relative times of acquisition. Taking into account the fact that the low-T_b component (probably

TABLE 2

Site mean directions: Torridonian components

Site No.	In situ		Dip corrected [a]		N	k	α_{95} (°)
	D (°)	I (°)	D (°)	I (°)			
(a) Low-T_b: SE + polarity							
1	118.1	33.9	120.2	48.7	3	13	36.4
2	129.7	66.6	157.3	79.5	3	36	21.0
3	174.7	77.8	240.9	75.3	2	–	–
4	133.5	36.7	139.9	50.1	4	19	21.7
6	144.9	51.3	160.6	62.4	2	–	–
7	128.1	34.5	132.8	48.6	2	–	–
10	153.4	41.8	165.6	51.6	2	–	–
12	142.4	30.8	149.0	43.0	2	–	–
Mean	135.1	46.7	145.3	59.7	20	11	10.5
directions	137.2	47.3	148.3	59.9	8	17	13.8
(b) High-T_b: SE + polarity							
4	124.8	21.3	127.0	35.7	2	–	–
5	144.1	37.6	153.0	49.4	1	–	–
10	141.6	50.2	155.7	62.0	1	–	–
13	124.9	43.6	130.5	57.9	1	–	–
17	129.0	56.8	142.2	70.4	1	–	–
Mean direction	131.3	39.0	137.7	52.6	6	20	15.2
(c) High-T_b: NW − polarity							
1	298.3	−21.2	299.6	−36.0	4	29	17.5
2	278.9	−7.7	278.1	−22.4	4	13	26.1
3	277.1	−48.9	270.9	−63.3	4	11	28.9
6	290.6	−28.1	290.7	−43.1	5	10	24.8
7	282.1	−40.9	279.4	−55.7	3	13	35.5
10	311.0	−18.5	313.7	−32.4	4	59	41.6
11	284.8	−21.1	284.0	−36.0	4	50	13.1
12	287.7	−31.9	287.1	−46.9	2	–	–
13	265.1	−25.8	260.7	−39.2	2	–	–
15	264.3	−26.4	259.7	−39.7	5	36	12.9
17	281.2	−19.9	279.9	−34.7	2	–	–
Mean	284.3	−26.8	283.2	−41.7	39	11	7.3
directions	284.0	−27.0	282.8	−41.9	11	24	9.5

Notes: (1) unit weighting to specimens; (2) unit weighting to sites.

[a] Tectonic correction of strike 200°, dip 15° appplied.

carried by magnetite) is never of multiple polarity and that magnetite would be very unlikely to form diagenetically in an open fluviatile system, it seems reasonable to conclude that it represents a detrital (or post-detrital) remanence [18].

The high-T_b remanence, however, may occasionally be of multiple polarity and individual components are generally better defined. It is therefore suggested that this represents the long-term acquisition of a chemical remanent magnetisation (CRM) over a period of time incorporating at least one field reversal. An estimate [12] of an average polarity interval of 1.5 m.y. in the Torridon Group as a whole suggests that the time scale of CRM acquisition in the Applecross Group of Rhum is of a comparable or greater duration.

Previous work using chemical demagnetisation [13] shows that the bulk of the stable remanence resides in detrital opaque grains rather than in the red pigment. Hence it is suggested that the CRM formed largely as a result of the partial oxidation of detrital magnetite grains during early diagenesis. The cores of such grains could remain unoxidised, continuing to record a magnetite DRM. Complete oxidation would tend to remove all such low-T_b remanence, as in specimen 1103 (Fig. 2c).

If such a model of remanence acquisition is applicable on a more regional scale throughout the Torridonian Sandstone, it could explain apparent incongruities in the magnetic stratigraphy record [13]. If the rate and degree of oxidation (martitisation?) was variable from area to area then in some localities the remanence would be dominated by diagenetic haematite, which would not necessarily record the true polarity at the time of deposition. On this basis, the only components which should be used to construct a valid magnetic stratigraphy would be those which can be shown to have been acquired during or very shortly after deposition.

5.2. *Tertiary*

As discussed above, Tertiary igneous rocks of Rhum are known to be predominantly of reversed polarity with the exception of a small number of dykes [14,15]. The mean direction from the former study of $D = 176.7°$, $I = -58.2°$ is used here as an aid to the identification of Tertiary components related to the Central Intrusion.

The occurrence of postulated Tertiary overprints in Torridonian sediments is listed in Table 3. The two sites within 1 km of the intrusion (sites 19 and 20) carry a Tertiary remanence only. The dominant remanence carrier in these sites is magnetite; however, in a few specimens where the

TABLE 3

Site mean directions: Tertiary components

Site No.	In situ		Dip corrected		N	k	α_{95} (°)	Distance from intrusion (km)	Tectonic correction	
	D (°)	I (°)	D (°)	I (°)					strike	dip
(a) Aureole of Central Intrusive										
20	179.2	−51.7	189.6	−24.4	4	42	14.4	0.50	300	30
19	184.5	−64.9	208.2	−34.9	6	166	5.2	0.95	140	35
18	190.8	−71.3	157.0	−64.4	6	43	3.2	1.45	200	15
17	165.4	−54.6	152.0	−44.6	5	15	6.2	2.10	200	15
15	186.7	−65.4	161.0	−58.6	3	11	11.6	2.50	200	15
13	147.7	−44.6	140.9	−32.1	3	15	32.8	2.90	200	15
12	184.5	−41.4	173.2	−36.0	2	–	–	3.10	200	15
11	197.8	−69.6	163.7	−64.4	3	64	15.6	3.35	200	15
10	205.1	−56.0	183.3	−54.4	3	9	44.3	3.50	200	15
Mean	179.7	−60.8	–	–	35	22	5.3			
directions	–	–	173.9	−48.7	35	11	7.6			
	180.1	−58.9	–	–	9	32	9.2			
	–	–	171.4	−47.9	9	15	13.9			
(b) Aureole of dyke										
16	024.7	63.5	–	–	6	35	11.5	2.40		

Notes: (1) unit weighting to specimens; (2) unit weighting to sites.

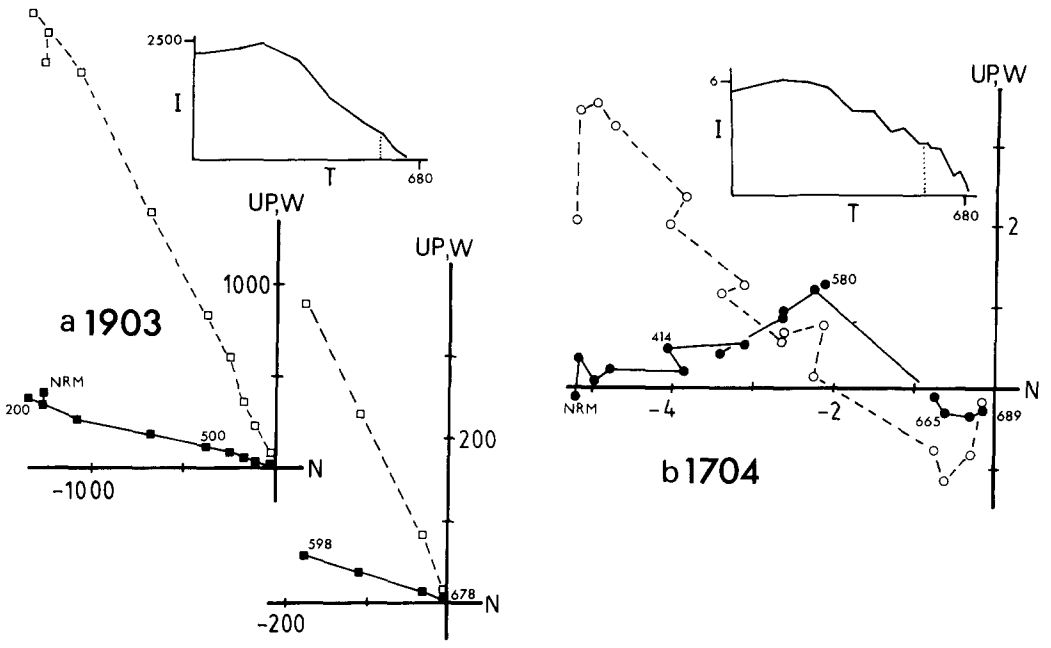


Fig. 3. Tertiary overprints: progressive thermal demagnetisation (details as for Fig. 2). Also shown are T_b spectra for each specimen, with remanent intensity I (in mA/m) plotted against demagnetisation temperature T (in ° C). (a) 1903 (0.95 km from the ring fault). Tertiary remanence up to the T_c of haematite. Inset shows high-temperature behaviour. (b) 1704 (2.1 km). Tertiary to 580 ° C followed by multi-component Torridonian.

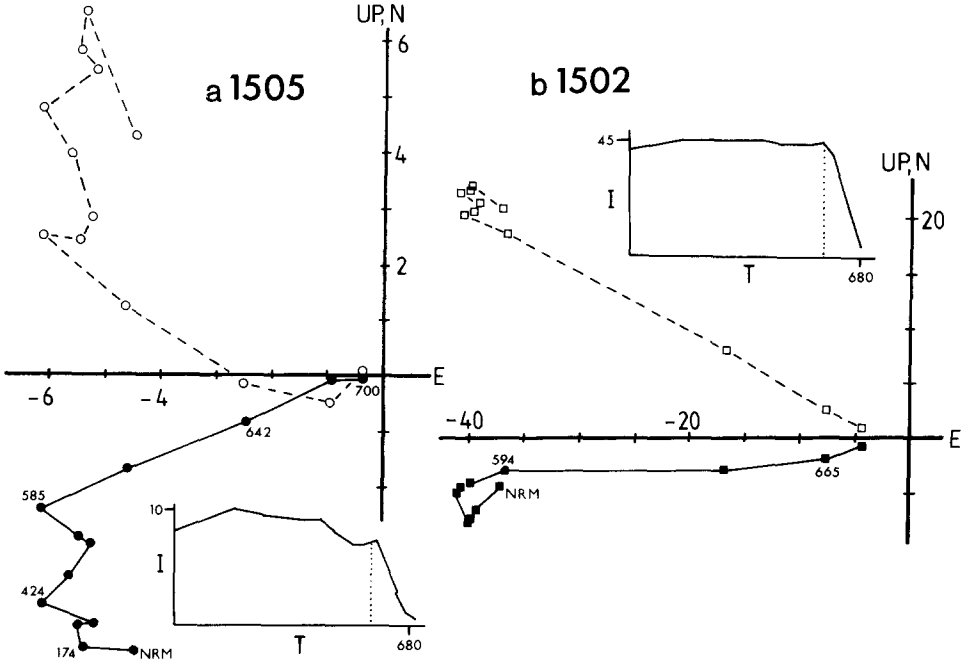


Fig. 4. Tertiary overprints: (a) 1505 (2.5 km). Tertiary to 585 ° C, followed by Torridonian NW−. (b) 1502 (2.5 km). Insufficient grains of low-T_b range to record a Tertiary overprint; high-T_b Torridonian NW− only.

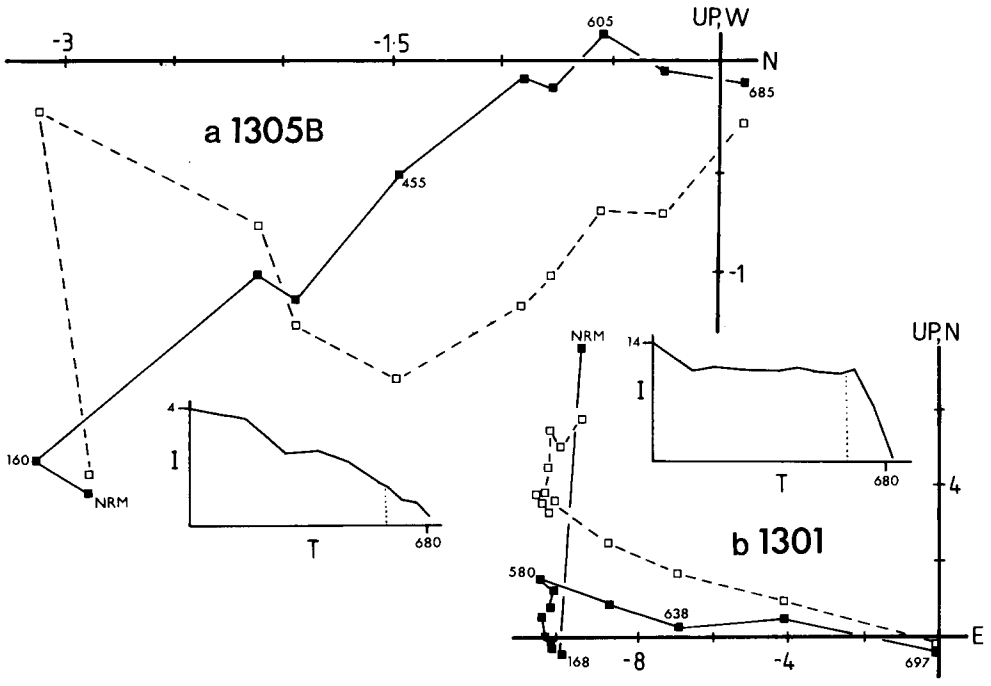


Fig. 5. Tertiary overprints: (a) 1305B (2.9 km). Tertiary 168–455 °C followed by Torridonian NW−. (b) 1301 (2.9 km). Tertiary 168–?580 °C followed by Torridonian NW−.

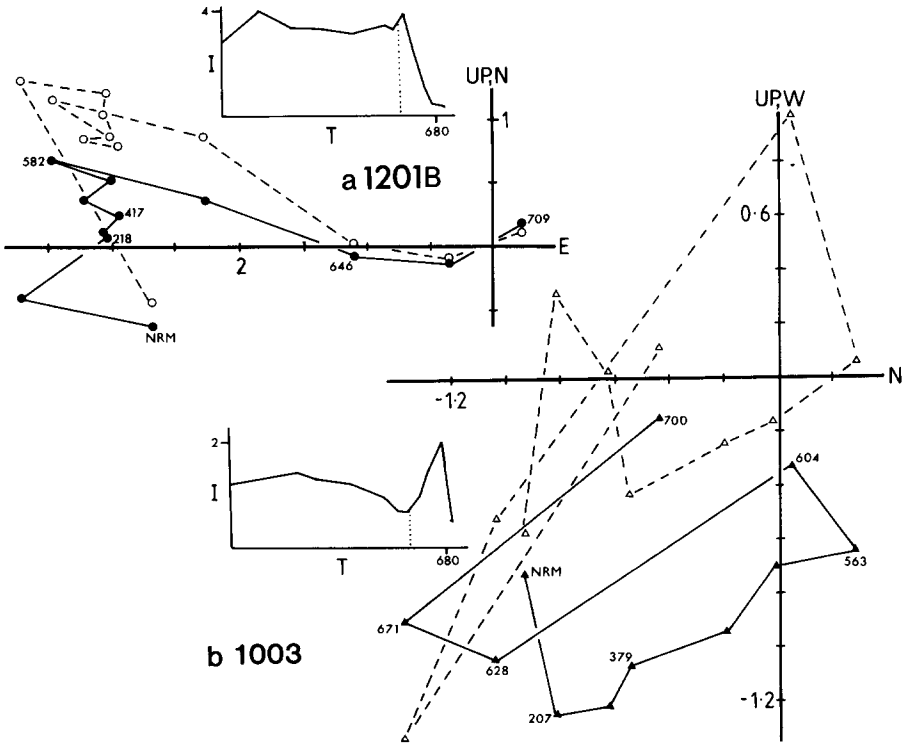


Fig. 6. Tertiary overprints: (a) 1201B (3.1 km). Tertiary to 417 °C, Torridonian SE+ 417–582 °C, Torridonian NW− 582–680 °C (b) 1003 (3.5 km). Tertiary 207–379 °C, Torridonian SE+ 379–563 °C, Torridonian NW− 604–628 °C, Torridonian SE+ 671–700 °C.

hematitic remanence is sufficiently intense, a Tertiary overprint can be detected up to the Curie Point of haematite (see specimen 1903, Fig. 3a).

Site 18 (1.45 km from the ring fault) also has a Tertiary overprint carried predominantly in magnetite but the Tertiary overprint in haematite has maximum blocking temperatures of approximately 620 °C. Above this a northerly downward component is revealed in three out of six specimens. This component is somewhat cryptic since it is not isolated in any other site.

All specimens from site 17 (2.1 km) have a Tertiary overprint with a maximum T_b of 580 °C. At this site a Torridonian remanence with maximum T_b greater than 580 °C has survived. 1704, for example, has a multi-component haematite Torridonian remanence (Fig. 3b).

Some specimens from site 15 (2.5 km) have a magnetite overprint with blocking temperatures up to 580 °C, but this is only observed when there are sufficient grains with blocking temperatures in this range. Other specimens from this site are dominated by a haematite Torridonian remanence and have a low-intensity Tertiary overprint which cannot be accurately defined as there is little material present of sufficiently low blocking temperature. Fig. 4a, b compares vector plots of thermal demagnetisation and demagnetisation spectra for specimens 1502 and 1504, which show such contrasting behaviour.

At distances of between 2.5 and 3.5 km from the ring dyke (sites 14 to 10), the occurrence of Tertiary overprints becomes patchy. This is partly because site 14 and some of site 13 have been severely affected by IRM, but it also reflects the change in T_b spectra which occurs as the distance from the intrusion increases. Specimens from those sites which have a low-T_b fraction tend to show a Tertiary overprint; Fig. 5a, b shows representative examples. The overprint here has a maximum T_b of 500 °C.

The remanence intensity generally decreases with distance from the intrusion (see below) and vector plots consequently tend to be less well-defined. Specimen 1201B (Fig. 6a) shows the removal of a Tertiary component with a maximum T_b of 420 °C, a SE+ Torridonian component between 420 ° and 580 °C and a NW component above 580 °C. Specimen 1003 shows the most complex behaviour of any seen in this study, with a Tertiary component between 200 ° and 380 °C, Torridonian SE+ between 380 ° and 580 °C, NW− from 580 ° to 670 °C and SE+ again from 670 ° to 700 °C. However, the low intensity and consequent high errors for this specimen suggest that this may be an over-interpretation of the data.

Specimen 1502 (Fig. 4b) is illustrated as an example showing no low-T_b Tertiary component. There is no evidence of any Tertiary overprinting in sites at greater distances from the ring fault than 3.5 km.

Bed dip correction. As previously mentioned, the two most proximal sites have a different tectonic dip to all other sites. Application of this correction to means for these sites results in a significant decrease in the precision parameter k (Table 3), suggesting that acquisition of the Tertiary remanence postdated deformation.

Susceptibility and NRM intensity distribution. There is a significant change in unit susceptibility of the Torridonian as the intrusion is approached (Table 4). This presumably reflects increasing magnetite content with proximity to the ring fault (magnetite having a much higher unit susceptibility than haematite). It is not appropriate to analyse the distribution of NRM intensity as this has frequently been distorted by IRM acquisition.

Dyke, site 16. Site 16, approximately 2.4 km from the ring fault, is from the baked margin of a 50 cm wide Tertiary dyke intruded into the Torridonian. Seven specimens were collected at distances of between 5 and 70 cm from the dyke margin. Apart from one suspected misoriented core, all specimens show a clear normal Tertiary direction (Fig. 7a, b); this is of opposite polarity to the main intrusion, although such directions have previously been found in a small number of dykes on Rhum [14]. NRM intensity and initial susceptibility both appear to increase significantly within 20–30 cm of the dyke (Fig. 7c), possibly in a comparable manner to that in the aureole of the main intrusion. No non-Tertiary high-T_b components are revealed during thermal demagnetisation, even though a significant proportion of the NRM intensity of each specimen survives well above the Curie point of magnetite, usually up to 660–680 °C.

TABLE 4

Variation of initial susceptibility with distance from the central intrusive

Site	Mean susceptibility
1–15	16.3
17	16.3
18	51.5
19	131.9
20	219.9

6. Origin of the Tertiary NRM and constraints on temperatures reached in the aureole

A number of workers [19–22] have used palaeomagnetism to determine temperatures reached by host rocks heated by igneous intrusions. If a rock is heated by such contact metamorphism to less than the Curie temperature of its magnetic constituents, then on cooling its remanence will consist of a low-T_b reset component and a high-T_b primary component. Progressive laboratory thermal demagnetization of this sample will remove all of the reset overprint and none of the primary remanence at a temperature equivalent to the temperature reached in the natural event. More details of this technique are given in [22]. For this method to be successful, it is necessary that either no chemical alteration of the remanence carriers occurred during the heating, or that if such alteration occurred, the CRM associated with the alteration can be separated magnetically from the thermal overprint [23]. This is because the blocking temperature of magnetic grains which carry CRM is not related to the temperature of formation of those grains but to their grain size and shape.

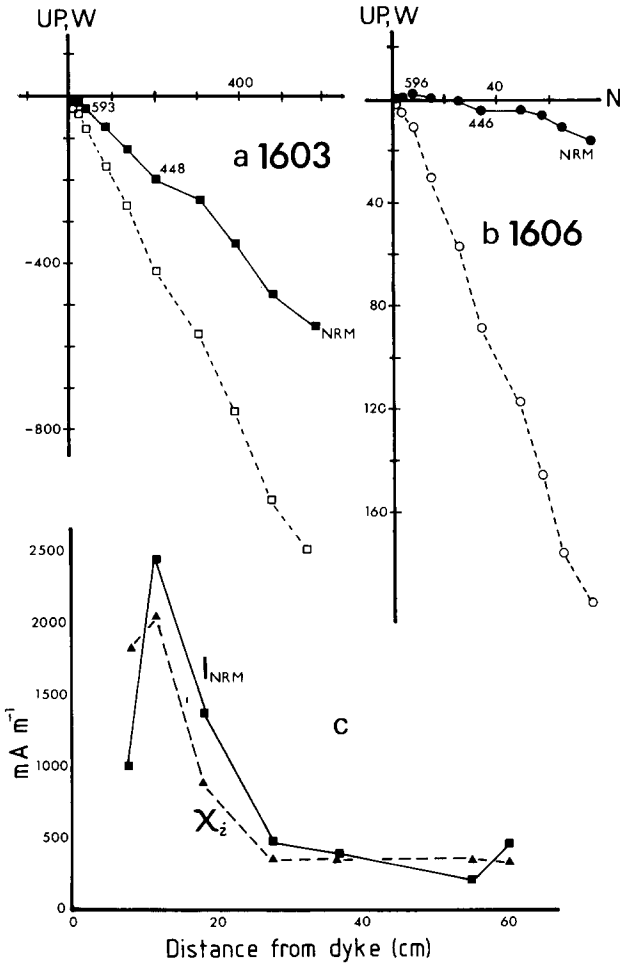


Fig. 7. Tertiary dyke margin, site 16. (a, b) Progressive thermal demagnetisation: details as for Fig. 2. (c) Variation of NRM intensity and initial susceptibility with distance from the dyke margin.

Within 2.1 km of the intrusion, the magnetite content of the sediments seems to have been enhanced by the intrusion event, since T_b spectra show magnetite to be the dominant remanence carrier and this is exclusively magnetised in a Tertiary direction. Remanent intensity and susceptibility are also significantly enhanced in this region (Table 4). Some specimens at site 15 (2.5 km) also carry a Tertiary overprint throughout the magnetite blocking range (up to 580 °C). Further than 2.5 km from the intrusion no apparent enhancement of the magnetite content is observed.

The pattern of Tertiary overprinting in haematite and magnetite described earlier suggests that there has been thermal overprinting as well as CRM acquisition by new magnetite introduced by the proximity of the intrusion. The maximum T_b of the thermal overprints should thus indicate the maximum temperatures reached in the contact aureole.

Fig. 5 shows the constraints on temperatures in the aureole. Close to the intrusion both haematite and magnetite have been remagnetized. It seems highly unlikely that a suitable chemical environment could exist which would cause the formation of both magnetite and haematite at the same time, so we postulate that the pre-existing haematite in this region has been thermally remagnetised and the magnetite has largely been newly formed, carrying a CRM.

A preliminary study by the first author of total iron content, measured using the inductively coupled plasma (ICP) spectroscopy method, shows that there is no significant increase towards the intrusion. This suggests that the magnetite has been formed by breakdown of pre-existing non-magnetic Fe-bearing minerals rather than by a net introduction via circulating fluids.

Sites 19 and 20 appear to have been heated above the haematite Curie point (about 680 °C) while at site 18 the Tertiary haematite has a maximum T_b of 620 °C. At distances of up to 2.5 km (site 15), magnetite carries a CRM; hence the Tertiary remanence has a maximum T_b up to the Curie point of magnetite. This can provide only a maximum temperature estimate for sites 17 and 15. At site 13 (2.9 km), the maximum T_b of the Tertiary overprint in specimen 1305B (Fig. 5a) is not well defined but lies in the range 400–500 °C. Elsewhere in the site, the Torridonian component has a minimum T_b of about 410 °C.

At site 12 (3.1 km), the maximum T_b of the Tertiary component in specimen 1201B is 320–420 °C and the minimum T_b of the lower blocking temperature Torridonian components is 420–520 °C. The minimum T_b of the Torridonian component in specimen 1203 is 330–380 °C; there may be a poorly defined Tertiary component below this.

At site 10 (3.5 km), the maximum T_b of the Tertiary component in specimen 1003 is 257–380 °C and the minimum T_b of the lower Torridonian component is 380–490 °C. In specimen 1005 the Tertiary component is poorly defined but

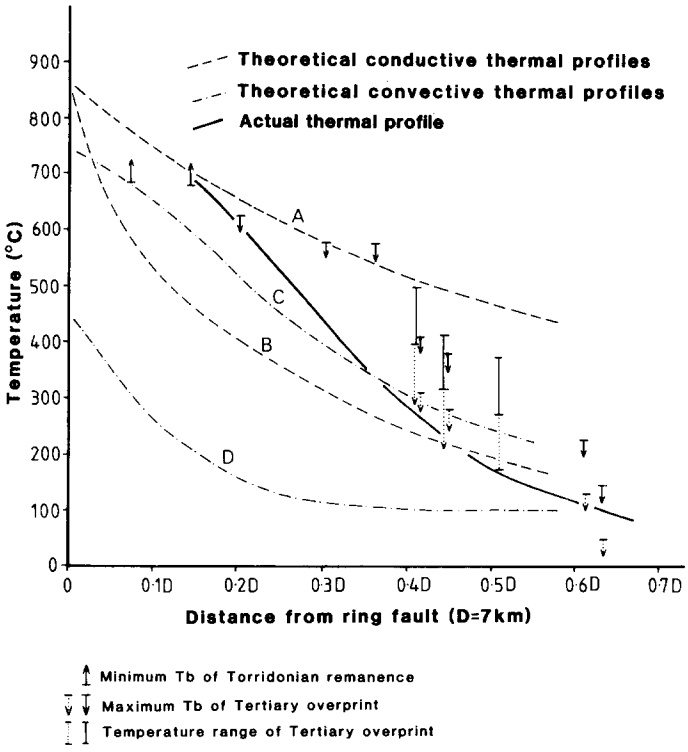


Fig. 8. Constraints on temperatures reached in the aureole of the Rhum central intrusive, according to distance from the ring fault. Distance is expressed as a function of diameter D of the central intrusive. Estimates of laboratory equivalents of the maximum temperatures reached expressed as solid vertical error bars; time-corrected equivalents shown (where appropriate) as dotted bars (see text). Solid line gives an estimated actual temperature profile in the aureole, based on experimental data given in this study. Also shown are calculated theoretical temperature profiles for various circumstances. *A*: conductive heat loss for vertical dyke, 4.5 km width; *B*: conductive heat loss for a vertical cylinder of 7 km diameter; *C*: convective heat loss for a 7 km slab: closed circulation, $Ra = 100$; *D*: convective heat loss for a 7 km slab: open circulation, $Ra = 50$. Contact temperature for curves *A* and *B* taken as 870 °C, corresponding to the quartz-tridymite transition (see text). Ambient temperature taken to be 100 °C, intrusion temperature as 1300 °C.

appears to have a maximum T_b of 280–350 °C while the minimum T_b of the Torridonian component is 580 °C. Further from the intrusion than 3.5 km Tertiary overprints are not seen. However, the minimum T_b of the Torridonian component can be used to provide an estimate of the maximum temperature reached. At site 6 (4.1 km) this is 240 °C while for site 5 (4.3 km) the minimum is 150 °C.

A further temperature constraint is provided by the existence of quartz pseudomorphs after tridymite in the sandstones inside and outside the ring fault ([5], and personal observation), since the temperature at which quartz converts to tridymite is 870 °C [24].

Together, these temperature constraints (illustrated in Fig. 8) provide an estimate of maximum temperatures reached in the contact aureole of the Rhum igneous centre. Since thermal remagnetisation is a kinetic process, temperatures estimated from experiments carried out over laboratory time scales will be higher than the actual temperature reached. This temperature profile can be compared to profiles obtained from thermal modelling calculations.

7. Thermal modelling

An igneous intrusion may cool by purely conductive heat transport if there is no hydrothermal fluid movement, or by convective heat transport if fluid movement is significant. Equations for conductive heat transport [25] and the maximum temperatures reached in a conductive aureole can be calculated for simple intrusion shapes, using numerical models.

The ultramafic central complex on Rhum is approximately cylindrical in shape with a diameter of about 7 km (see Fig. 1). The igneous complex as a whole is elongated in the east-west direction. The simplest shape which can be modelled is a dyke-like body of infinite vertical and horizontal extent. A width of 4.5 km has been chosen to roughly fit the shape of the intrusion. Calculations of heat transport in the contact zone of a dyke have previously been published [26], although details of the calculations are not given.

The temperature in the neighbourhood of a vertical sheet can be recalculated to allow for the latent heat of crystallisation in the intrusion [27]. An analytic solution for the temperatures in two semi-infinite bodies in contact at $x = 0$ was determined at small values of time. At short times after the intrusion, the temperature at the centre of the intrusion is still equal to the intrusion temperature and the analytic approximations are valid. Once the centre cools below the intrusion temperature, a "reflection" of the heat loss at the other boundary begins to affect the temperature in the central region. So the temperature in the intrusion and in the aureole were calculated numerically using difference methods at a large number of increments of time using the analytic solution at a small value of time as a starting solution. It is not desirable to only use a numerical solution since the initial temperature condition of a step function across the contact causes inaccuracies in the difference computations.

A numerical solution to the heat flow in the contact of a cylindrical intrusion has been given by [28]. The intrusion temperature of the Rhum ultrabasic intrusion has been estimated to be approximately 1300 °C [29]. For a single pulse of magma intruded into country rock at an ambient temperature of 100 °C, the calculated contact temperature would be approximately 820 °C. However, there have been at least 15 batches of magma injected into the layered basic intrusion; the magma chamber was probably cyclically refilled over a period of a few thousand years (Palacz, personal communication). To make an allowance for this cyclic replenishment of the magma chamber, we have used an arbitrary contact temperature of 870 °C (the temperature of the tridymite/quartz transition, since tridymite is found in the Torridonian Sandstone immediately adjacent to the intrusion: see above). Maximum temperature profiles in the aureole of a conductively cooled intrusion calculated for a vertical sheet of width 4.5 km or for a vertical cylinder of diameter 7 km, using the above parameters, are also shown in Fig. 8. The width of the aureole is strongly controlled by the intrusion geometry.

Peak temperatures in contact aureoles for convectively cooled intrusions can be calculated [28]. A variety of systems were considered, with either permeable or impermeable intrusions, closed hydrothermal circulation with no escape of fluid at the surface, open circulation with surface fluid loss and two values of country rock permeability.

$\delta^{18}O$ values [9] show that some hydrothermal fluids passed through the Rhum intrusive, so only the solutions for permeable intrusions will be considered here. The effect of surface loss of fluid in an open circulation system is to narrow the thermal aureole; a more permeable country rock widens the aureole. Values of maximum temperatures taken from graphs given by [28] for intrusion temperatures of 1300 °C and ambient temperatures of 100 °C are shown in Fig. 5 for comparison with conductive temperature profiles and the temperature estimates obtained from this study.

Conductive heat transport calculations indicate that rocks at 3 km distance from the intrusion would have experienced a thermal pulse of approximately 1000 years duration. The kinetic nature of the blocking process [30,31] means that a remanence unblocked in a laboratory experiment between 300 ° and 500 °C would have been unblocked in nature over a thousand year period about 100 °C below the laboratory estimate. More accurate corrections could be calculated but the palaeomagnetic data are not precise enough to warrant this. Temperatures which have been approximately corrected for the time difference are shown as dotted bars in Fig. 5, and a maximum temperature curve has been estimated for the experimental data.

The calculated temperature curve which fits these data best is that for convective heat transport in a closed hydrothermal circulation system around a 7 km wide slab with a country rock permeability of about 0.2 millidarcies. However, this model does not predict high enough temperatures close to the intrusion and is not physically acceptable, since the Torridonian sandstones are highly impermeable, probably having a permeability of less than a microdarcy (Lumsden, personal communication). The $\delta^{18}O$ data shows that hydrothermal circulation was restricted to a narrow region around the ring fault.

The experimental data lie within the envelope of the conductive heat transport profiles for a vertical dyke width 4.5 km and a vertical cylinder width 7 km. The dyke temperature profile is too hot far from the intrusion while the cylinder profile is too cool close to the intrusion. The ultrabasic complex is obviously not simple in shape and has associated outlying intrusions (see Fig. 1). The first 2 km of the palaeomagnetic sampling profile passed within a kilometer of two of these intrusions which may be more extensive at depth.

We therefore suggest that estimates of temperature in the thermal aureole described in this paper indicate that the approximately cylindrical central ultrabasic intrusion cooled conductively at a level similar to its present level with respect to the Torridonian, and that the temperatures close to the intrusion have been enhanced by the proximity of associated minor intrusions.

8. Conclusions

The high temperatures of greater than 700 °C obtained in the Torridonian sediments within 1 km of the ring fault indicate that the ultrabasic intrusion was emplaced at about 1300 °C at a position close to its present stratigraphic level. It has been suggested [5] that the ultrabasics were intruded at a lower level (perhaps as much as 2 km lower) and that they were emplaced in an essentially solid condition along the main ring fracture. The evidence presented in this paper indicates that any upward movement of the central complex was probably minor. Tongues of the ultrabasic complex cut the northern part of the ring fault, which supports our hypothesis. Temperature estimates obtained in the contact aureole by palaeomagnetic methods suggest that the intrusion cooled essentially by conductive heat transport.

Palaeomagnetic study of unheated Torridonian sandstones shows that the remanence may be carried by both magnetite and/or haematite. We believe that the magnetite remanence represents the original magnetisation direction recorded by detrital grains, whereas the haematite represents a CRM acquired later over a period of a million years or more.

Remanence in redbeds can often have such multi-component nature. If redbeds are to be used for magnetostratigraphic studies, it is therefore necessary to carefully assess which remanence component is truly primary and only this component should be used to construct a magnetostratigraphy.

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